Bishop's University

CS 596 – Special Topics on Deep Learning

Final project: Exploring Recurrent Neural Networks (RNNs) with PyTorch

A. Objective

The objective of this assignment is to gain hands-on experience with Recurrent Neural Networks (RNNs) using PyTorch. You will implement and train an RNN model to perform a specific task and explore its capabilities in sequence modeling.

B. Tasks

1. Background Research

Begin by researching the fundamentals of RNNs, their architecture, and applications in various fields such as natural language processing, time series analysis, and speech recognition. Familiarize yourself with PyTorch, a popular deep learning framework.

2. Setup Environment

Install PyTorch and any necessary dependencies on your local machine or a suitable environment. Ensure that you have access to Jupyter Notebook or any preferred Python IDE for coding.

3. Dataset Selection

Choose a suitable dataset for your RNN task. You can consider datasets related to text data, time series, or any sequential data that aligns with your interests. Popular datasets include text corpora, stock prices, weather data, etc.

4. Data Preprocessing

Preprocess the selected dataset to prepare it for training the RNN model. This may involve tasks such as tokenization, normalization, sequence padding, etc., depending on the nature of the data.

5. Model Implementation

Implement an RNN model using PyTorch. You can choose between different RNN variants such as Vanilla RNN, Long Short-Term Memory (LSTM), or Gated Recurrent Unit (GRU). Design the model architecture, including the input/output dimensions, hidden layer size, activation functions, etc.

6. **Training and Evaluation**

Train the RNN model on the preprocessed dataset using suitable training parameters such as batch size, learning rate, and number of epochs. Monitor the training process and evaluate the model's performance using appropriate metrics such as accuracy, loss, or any domain-specific evaluation metric.

7. **Analysis and Interpretation**

Analyze the results of your trained RNN model. Interpret its performance on the chosen task and identify any areas for improvement or further experimentation. Explore the model's predictions and understand its behavior in different scenarios.

8. **Documentation**

Document your entire workflow, including dataset details, preprocessing steps, model architecture, training process, evaluation results, and analysis. Provide clear explanations and code comments to ensure readability and reproducibility.

C. Deliverables:

- Jupyter Notebook or Python script containing the implementation of the RNN model,
- Documentation report summarizing the final project tasks, methodology, results, and analysis,
- Visualizations or plots illustrating key findings or insights from the experiment,
- Presentation slides summarizing key findings and insights for presentation purposes.

D. Submission Guidelines:

Submit your final project on Moodle. Ensure that all required deliverables are included and organized in a clear and structured manner. If applicable, provide instructions for reproducing your experiments or running your code.